Digital Twin for Indigo Dyeing

Autonomous Chemical Control & EU-Compliance Automation

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The Hidden Costs of Manual Dyeing

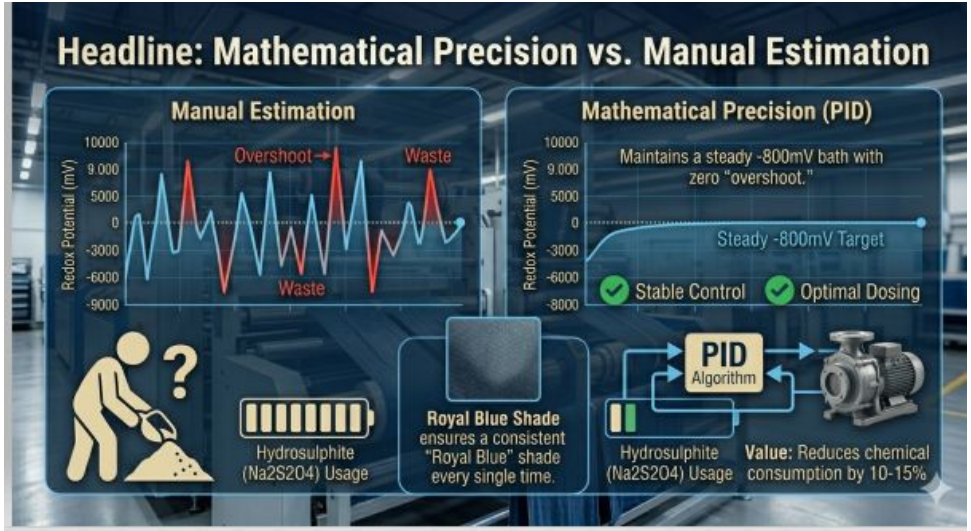
Chemical Waste: Manual "guesswork" leads to over-saturation of Hydrosulphite
Shade Variation: "B-Grade" fabric caused by human error during night shifts.
Export Barriers: New EU "Green Deal" regulations (2026) require Digital Product Passports (DPP) that manual mills cannot provide.
Mechanical Downtime: Sudden pump failures stop production for hours.

A 24/7 Autonomous "Expert" in Your Dye House

- **Multimodal AI:** A system that "sees" with cameras and "feels" with chemical sensors.
- **Closed-Loop Logic:** If the shade fades, the AI automatically corrects the chemistry in milliseconds—no human intervention required.
- **Predictive Intelligence:** Monitors machine vibrations to fix pumps *before* they break.

Mathematical Precision vs. Manual Estimation

PID Control: Uses the "Gold Standard" of chemical engineering to stabilize Redox potential.
Result: Maintains a steady -800mV bath with zero "overshoot."
Value: Reduces chemical consumption by **10-15%** and ensures a consistent "Royal Blue" shade every single time.



Automating EU-Market Access

Digital Product Passport (DPP): Every meter of fabric is automatically logged into a secure digital ledger.

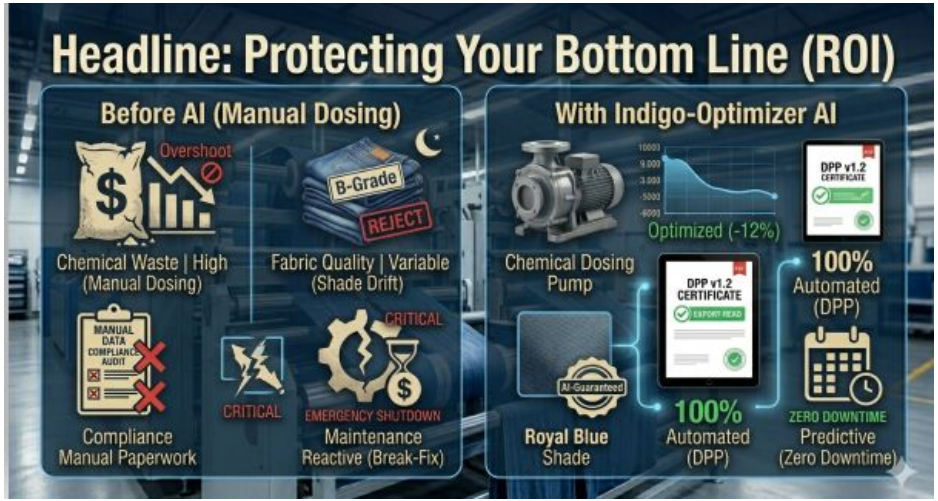
Sustainability Certificate: Generate a "Green Audit" at the click of a button.

Value: Instantly satisfy audits from global brands (Levi's, H&M, Zara) regarding chemical and water footprints.

Protecting Your Bottom Line

Chemical Waste | High (Manual Dosing) | **Optimized (-12%)**|

Fabric Quality | Variable (Shade Drift) | **Consistent (AI-Guaranteed)**|

Compliance | Manual Paperwork | **100% Automated (DPP)**

| **Maintenance** | Reactive (Break-Fix) | **Predictive (Zero Downtime)**